

Executive Business Report

Retail Sales Analysis: Forecasting, Anomalies & Demand Segmentation

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Executive Summary

We analyzed four years of company sales data to understand what has been driving performance and to predict what comes next. Sales have grown steadily year over year, with predictable seasonal peaks around major shopping periods. Using this data, we built a forecasting model that predicts strong sales in January, followed by a normal seasonal dip in February and March. We also identified a small number of unusual sales weeks that are worth a closer look, and grouped our products into four demand categories to guide smarter stocking decisions. Overall, the data points to two clear priorities: lean into the Technology category and the West region, where growth is strongest, and adopt a segment-based approach to inventory so we stock the right products in the right quantities.

Key Findings from Data Analysis & Forecasting

- Sales show a consistent upward trend from 2015 to 2018, with a repeating seasonal pattern each year — confirmed statistically as a stable, predictable pattern rather than random noise.
- We tested three forecasting approaches (SARIMA, Prophet, and XGBoost). Prophet produced the most accurate forecasts, with an average prediction error about 35% lower than SARIMA and 26% lower than XGBoost, so it was selected as the model going forward.
- Broken down by category, Technology is forecast to lead in sales, ahead of Office Supplies and Furniture. Broken down by region, the West is forecast to outperform the East in most upcoming months.

3-Month Sales Forecast

Based on the Prophet model, here is what we expect for the next three months, along with a realistic low-to-high range to account for normal uncertainty:

Month	Expected Sales	Likely Range (Low – High)
Month 1 (Jan)	\$106,500	\$97,100 – \$116,500
Month 2 (Feb)	\$44,100	\$35,000 – \$53,000
Month 3 (Mar)	\$40,400	\$31,600 – \$49,700

In plain terms: Month 1 is expected to be a strong month, likely driven by end-of-year and new-year order activity, followed by a normal seasonal cool-down in Months 2 and 3.

Top 3 Anomalies Detected

Week	Sales	Likely Cause
Mar 22, 2015	\$37,700 (spike)	Large bulk order, corporate purchase, or spring promotion
Dec 2, 2018	\$36,000 (spike)	Year-end / holiday season buying surge
Nov 18, 2018	\$30,600 (spike)	Holiday discounts or seasonal promotional campaign

These three weeks stood out the most from normal sales patterns. Sixteen unusual weeks were flagged in total across two detection methods; these three were the largest by dollar impact.

Product Demand Segmentation & Stocking Strategy

Products were grouped into four demand segments based on sales volume, growth, and stability:

Segment	Example Products	Recommended Stocking Strategy
High Value	Copiers, Machines	Keep adequate stock; monitor closely given high unit cost and volatility
High Volume, Stable	Chairs, Phones, Storage, Binders	Maintain higher stock levels with regular replenishment to avoid stockouts
Low Volume, Stable	Lower-selling steady items	Keep lean inventory; restock as needed to save on storage costs
Growing Demand	Supplies	Increase stock gradually while monitoring the upward trend

Business Recommendations

- 1. Prioritize the Technology category and the West region.** Forecasts show Technology leading next month's sales at roughly \$46,200, ahead of Office Supplies (\$30,000) and Furniture (\$28,800); the West region is forecast at \$38,800 versus \$31,900 for the East. Marketing and inventory budgets should weight toward these two areas.
- 2. Standardize on the Prophet model for ongoing forecasting.** Across three tested models, Prophet had the lowest average forecast error (about 14–15% off actual sales, versus 23%+ for SARIMA), making it the most reliable choice for planning purchasing and staffing.
- 3. Move to segment-based inventory planning.** Treat High Volume/Stable products (Chairs, Phones, Storage, Binders) as always-in-stock essentials, while gradually scaling up Supplies inventory to keep pace with its strong growth trend, reducing both stockouts and excess holding costs.

Risk & Limitation

This forecasting system is built on roughly four years of historical sales data and statistical pattern-matching. It assumes future conditions will broadly resemble the past, so it cannot anticipate one-off shocks such as supply chain disruptions, new competitors, or major economic shifts. Similarly, the anomaly detection flags statistically unusual weeks but does not confirm their actual cause — the explanations above are informed estimates and should be verified against real business records (e.g., promotions calendar, large order logs) before being used to justify major decisions.